

ROSE–HULMAN INSTITUTE OF TECHNOLOGY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

Final Report for

IoT Connected SmartHub Dashboard

ECE436 – Internet of Things

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May 15th, 2026

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1 Executive Summary

The IoT Connected SmartHub Dashboard is a multi-functional smart desk system developed using ESP32 microcontrollers in combination with LoRa, Bluetooth Low Energy (BLE), and Wi-Fi communication technologies. The system provides real-time weather updates, music playback, battery status, environmental sensor data, and nearby BLE devices through an interactive 7-inch dashboard display.

LoRa communication enables reliable long-distance wireless sensor data transmission, while BLE supports nearby device scanning and smartwatch interaction. Wi-Fi is used for web server access, remote interaction, and Apple Shortcut battery updates. The project also integrates a camera-based hand-gesture control system using Edge Impulse machine learning for automatic screen rotation and user interaction. A 7-inch display is connected to the main SmartHub device, where all real-time data and system information are visualized. A heavy-duty servo motor is used to rotate the display between 0° and 90° orientations for portrait and landscape viewing modes.

Overall, the SmartHub combines multiple IoT technologies into a single user-friendly dashboard platform designed to provide students and everyday users with convenient access to important real-time information from one centralized desk device.

2 Introduction

The Internet of Things (IoT) allows devices to connect, communicate, and share data automatically through the internet and wireless communication technologies. IoT systems improve convenience, automation, monitoring, and real-time interaction between smart devices. This project was developed to provide students with easier access to multiple smart functions through a single centralized dashboard system.

The IoT Connected SmartHub Dashboard is a multi-functional smart desk system developed using two Heltec WiFi LoRa ESP32 development boards and several additional ESP32-based components. The system integrates multiple communication technologies including LoRa for long-distance communication, Bluetooth Low Energy (BLE) for short-range device interaction, and Wi-Fi for web server access and remote communication.

The SmartHub supports real-time information such as weather updates, Spotify music playback, battery status, environmental sensor data, and nearby BLE device detection. The system also includes a 7-inch ESP32-based display used to visualize dashboard information and user interaction. The display supports both portrait and landscape orientations through a heavy-duty servo motor system. With the help of camera-based gesture recognition, users can rotate the screen with simple hand gestures.

The overall goal of the project is to combine multiple IoT technologies into a single user-friendly

SmartHub device designed for everyday student use.

3 Requirements and Specifications

3.1 Functional Requirements

- The SmartHub shall display real-time weather information.
- The system shall display Spotify music playback information.
- The system shall display environmental sensor data including temperature and humidity.
- The SmartHub shall support nearby BLE device scanning and detection.
- The system shall support long-distance LoRa communication between ESP32 nodes.
- The SmartHub shall provide Wi-Fi web server access for remote interaction and monitoring.
- The system shall support camera-based hand gesture recognition.
- The SmartHub shall support automatic screen rotation using a servo motor system.
- The dashboard shall support both portrait and landscape display orientations.
- The system shall support real-time updates for dashboard information and sensor values.

3.2 Non-Functional Requirements

- The system shall provide reliable wireless communication between subsystems.
- The dashboard interface shall remain user-friendly and responsive during operation.
- The LoRa transmitter node shall support low-power operation using deep sleep functionality.
- The system shall maintain stable operation during continuous runtime.
- The project shall support modular subsystem integration for easier debugging and maintenance.
- The system shall support scalable firmware development using PlatformIO environments.
- The SmartHub shall support smooth user interaction and display rendering using LVGL.
- The system shall maintain reliable servo motor operation without interrupting communication subsystems.

4 Research and Current Work Review

The SmartHub Dashboard project started with individual subsystem testing to ensure that each component operated correctly and fulfilled its intended purpose. After completing subsystem-level testing, the project progressed toward full system integration using two Heltec ESP32 Wi-Fi LoRa development boards and multiple wireless communication technologies. To complete the physical assembly of the system, a custom housing structure was implemented to support the main SmartHub ESP32 board, display stand, and camera holder. After completing the wiring and hardware integration, the entire system was tested to verify proper power distribution, subsystem communication, and overall functionality.

Current development focuses on integrating LoRa communication, BLE scanning, Wi-Fi web server functionality, environmental sensing, display rendering, gesture recognition, and power analysis for the overall SmartHub architecture.

The system currently uses a 7-inch Elecrow ESP32-S3 display running LVGL and LovyanGFX for the main dashboard interface. BLE functionality has been implemented for nearby device scanning and smartwatch communication, while Wi-Fi supports web server access and Apple Shortcut battery updates for iPhone and iPad devices. LoRa communication between ESP32 nodes has been successfully tested using 915 MHz SX1262 transceivers (TX and RX) for reliable long-distance environmental sensor data transmission.

The project also integrates a camera-based hand gesture recognition subsystem using the Seeed Studio XIAO ESP32S3 Sense module and Edge Impulse machine learning. The gesture recognition system currently supports directional gesture detection with approximately 92.9% accuracy.

Additional current work includes subsystem integration, automatic brightness control, power optimization, battery lifetime estimation, and final system debugging to improve overall system reliability and user experience.

5 Design Method and Overview

The SmartHub Dashboard was designed using a modular IoT architecture based on multiple ESP32 microcontrollers and wireless communication technologies. The system integrates LoRa, Bluetooth Low Energy (BLE), and Wi-Fi communication into a centralized dashboard platform for real-time monitoring, interaction, and device control. The very first architecture design is shown in Figure 2, while the final version of the architecture system for the SmartHub is displayed in Figure 1.

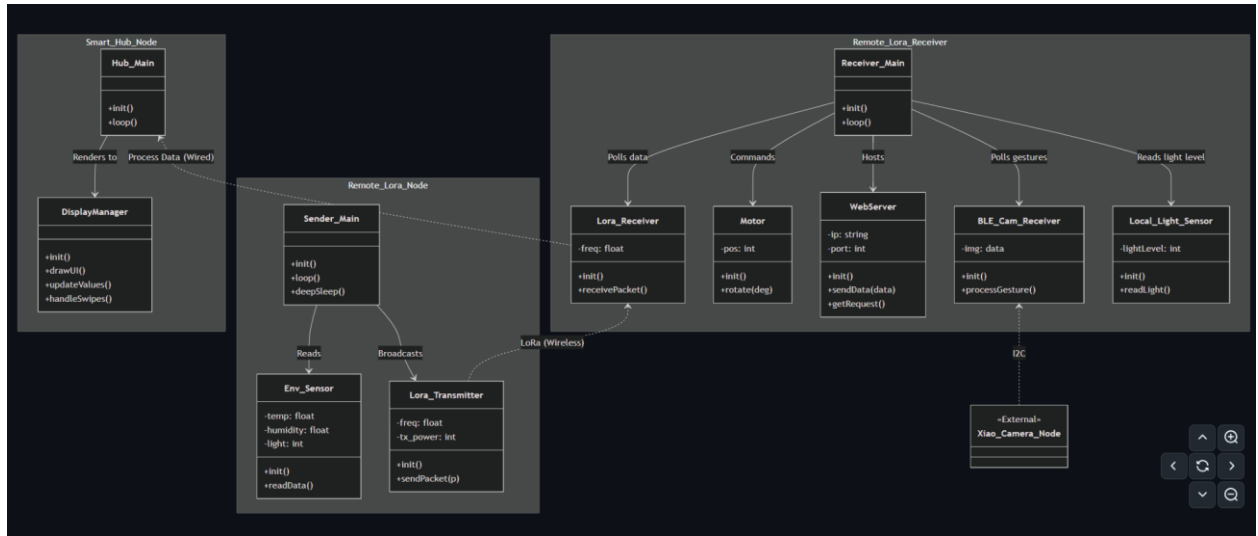


Figure 1. Overall smartHub Architecture.

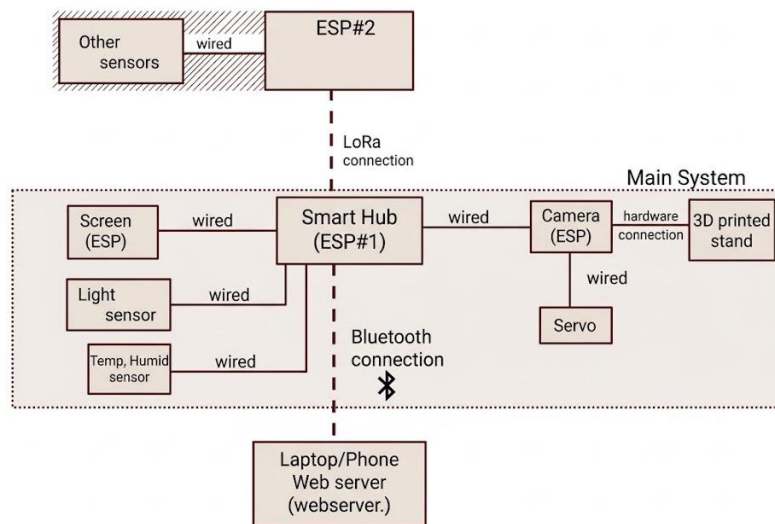


Figure 2. The first architecture design for the overall system.

The main SmartHub ESP32 acts as the central controller and connects to multiple subsystems including the 7-inch Elecrow ESP32-S3 display, LoRa receiver node, environmental sensors, web server interface, speaker system, and camera-based gesture recognition module. The dashboard interface was developed using LVGL and LovyanGFX to provide a responsive and user-friendly graphical display system.

LoRa communication is implemented using Heltec ESP32 LoRa development boards operating at 915 MHz with SX1262 transceivers for reliable long-distance environmental data transmission. BLE functionality is used for nearby device scanning and smartwatch communication, while Wi-Fi

provides web server access, remote interaction, and Apple Shortcut battery updates.

The system also integrates a camera-based gesture recognition subsystem using the Seeed Studio XIAO ESP32S3 Sense module and Edge Impulse machine learning. Gesture detection is used for automatic screen rotation and dashboard interaction through directional hand gestures.

The overall design approach focuses on modular subsystem integration, reliable wireless communication, low-power operation for remote sensor nodes, and a user-friendly dashboard interface for everyday use.

6 Team Member Contributions and Work Completed

6.1 Anar Amartuvshin

- Integrated BLE and Wi-Fi subsystems into the SmartHub dashboard architecture.
- Implemented nearby BLE device scanning and RSSI monitoring functionality.
- Developed Apple Shortcut battery update communication for iPhone and iPad devices through Wi-Fi.
- Designed and integrated the “Nearby Devices” dashboard interface.
- Assisted with camera dataset collection and gesture labeling for machine learning training.
- Designed the smartHub closure.

6.2 Johannes Hirt

- Integrated the LoRa receiver subsystem into the SmartHub architecture.
- Implemented LoRa packet reception, parsing, RSSI monitoring, and SNR measurement.
- Developed the SmartHub web server and remote-control interface.
- Integrated environmental sensing and automatic brightness control systems.
- Performed Packet Delivery Ratio (PDR) testing and received power analysis.
- Implemented and tested the servo motor subsystem.
- Assisted with camera dataset collection and gesture labeling.

6.3 Thiago Costa

- Developed the LVGL-based SmartHub dashboard interface for the 7-inch display.
- Implemented multiple dashboard pages including weather, battery, music playback, and BLE device visualization.
- Replaced the original OLED implementation with the larger touchscreen interface.
- Trained and optimized the Edge Impulse hand gesture recognition model.
- Improved camera gesture recognition performance and machine learning integration.
- Assisted with overall subsystem integration and UI improvements.

6.4 Overall Work Completed

- Full SmartHub dashboard integration using multiple ESP32 devices.
- Successful LoRa communication between ESP32 nodes at 915 MHz.
- BLE nearby device scanning and smartwatch interaction.
- Wi-Fi web server communication and battery update system.
- Camera-based hand gesture recognition with approximately 92.9% accuracy.
- Servo motor integration for automatic screen rotation.
- Environmental sensing and automatic brightness control.
- Power analysis and battery lifetime estimation for multiple subsystems.
- Real-time dashboard visualization using LVGL and LovyanGFX.

7 Verification and Results

The SmartHub Dashboard subsystems were verified through integration and communication testing across LoRa, BLE, Wi-Fi, display rendering, environmental sensing, and gesture recognition systems. Testing confirmed successful real-time communication between ESP32 nodes and stable dashboard operation.

LoRa communication testing demonstrated reliable packet transmission and reception at distances up to 100 m under free-space conditions. Packet Delivery Ratio (PDR) measurements (Figure 3) showed highly reliable communication performance, while received power measurements (Figure 4) followed the expected logarithmic path loss behavior.

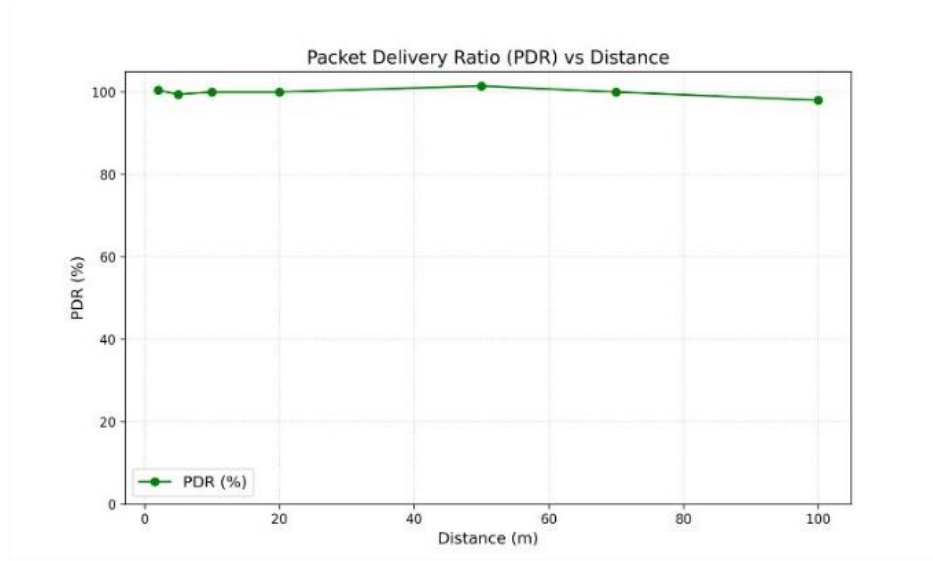


Figure 3. Packet Delivery Ratio (PDR) vs. distance for the 915 MHz LoRa link.

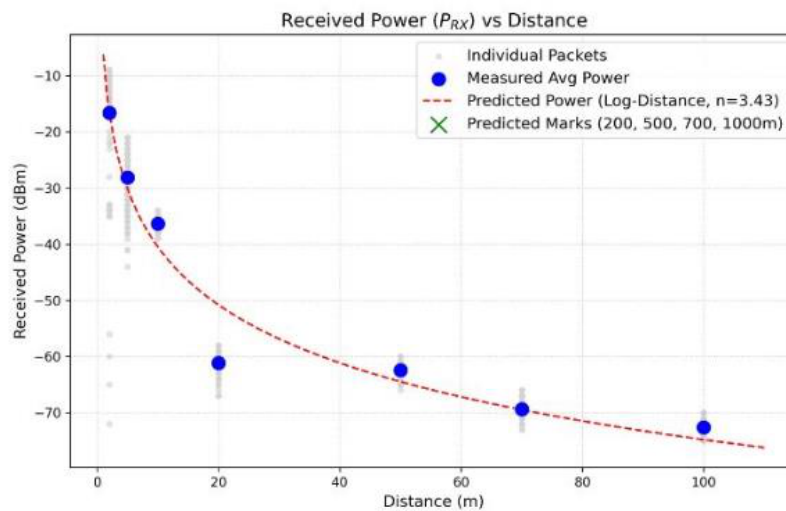


Figure 4. Received power vs. distance, with a log-distance path loss fit ($n = 3.43$).

BLE testing confirmed successful nearby device scanning, RSSI monitoring, and smartwatch detection. Wi-Fi testing verified stable web server communication and successful Apple Shortcut battery updates for iPhone and iPad devices.

The display subsystem successfully rendered dashboard pages including weather information, battery status, environmental sensor values, and nearby BLE devices using the 7-inch ESP32-S3 display. Automatic brightness adjustment and portrait/landscape display modes were also successfully tested.

Gesture recognition testing using the ESP32-S3 camera module and Edge Impulse achieved approx-

imately 92.9 % accuracy for directional hand gestures including LEFT, RIGHT, UP, and DOWN.

Power analysis results confirmed stable subsystem operation and demonstrated the effectiveness of separating the system into multiple isolated power domains for improved reliability and reduced power-related communication interruptions.

8 Power and Lifetime Analysis

To determine the realistic power requirements and to estimate the battery lifetime for each subsystem, we performed a measurement-based power analysis. Current draw was captured for the three independent power domains in the system: the central SmartHub receiver, the isolated servo motor, and the remote LoRa transmitter node.

8.1 Measured Power Sources

The IoT SmartHub system is partitioned into three power domains, each with its own supply. Measured current draws are summarized below.

Central SmartHub (Receiver ESP32 & Display). Built from a Heltec WiFi LoRa 32 V3, a 7-inch display, a continuous Wi-Fi connection, and the web server. Measured current is ~ 470 mA (0.47 A) active continuous. Powered from wall power via USB-C or a dedicated 5 V/3 A+ supply – driving a 7-inch backlight and maintaining continuous Wi-Fi/web server connectivity drains too much current for a battery to be practical.

Actuator (Servo Motor). A 20 kg RC servo motor. Measured current is a peak of 8.2 A, an average of 2.44 A (2440 mA) while active, and 0.06 mA at idle. Powered from a separate 5 V to 7 V battery pack (e.g., a standard 2000 mA h receiver pack). Isolating the servo prevents its 8.2 A peak spikes from browning out or resetting the ESP32 microcontroller.

Remote Environment Sensor (Transmitter Node). A Heltec WiFi LoRa 32 V3 paired with a DHT11/BHT11 environment sensor. Measured current is 0.5 mA during idle/deep sleep and 129 mA (0.129 A) while transmitting. Powered from a 3.7 V LiPo battery (e.g., 2000 mA h) through the Heltec onboard battery management circuit. This node reads simple sensor data and broadcasts it via LoRa, so it can leverage deep sleep between readings to maximize battery life.

8.2 Transmitter Node Average Current

To model the power usage of the battery-operated transmitter node, we divide its operation into two states – *active* and *idle/deep sleep* – and weight each state by the fraction of the update interval it occupies.

Parameters:

- Idle current $I_{\text{sleep}} = 0.5 \text{ mA}$
- Active (transmitting) current $I_{\text{active}} = 129 \text{ mA}$
- Active duration $T_{\text{active}} \approx 2 \text{ s}$
- Update interval $T_{\text{interval}} = 5 \text{ min} = 300 \text{ s}$

$$\begin{aligned}
 I_{\text{avg_tx}} &= \frac{I_{\text{active}} T_{\text{active}} + I_{\text{sleep}} (T_{\text{interval}} - T_{\text{active}})}{T_{\text{interval}}} \\
 &= \frac{(129 \text{ mA})(2 \text{ s}) + (0.5 \text{ mA})(298 \text{ s})}{300 \text{ s}} \\
 &= \frac{258 + 149}{300} \text{ mA} = \frac{407}{300} \text{ mA} \approx 1.356 \text{ mA}.
 \end{aligned}$$

8.3 Receiver Node (SmartHub)

The central hub remains on continuous wall power due to its constant high-current requirement. No averaging is needed because the load is effectively a constant base current:

$$I_{\text{base}} = 470 \text{ mA}.$$

Running this load from a 2000 mA h battery would exhaust the usable capacity in roughly $1700/470 \approx 3.6 \text{ h}$, which is why the receiver is permanently wall-powered.

8.4 Servo Motor Average Current

The servo is powered by an independent 5 V to 7 V battery to isolate its current spikes from the main receiver. Its power usage is modeled around an hourly sweep, with the idle term retained because it represents the standing current of the isolated supply rail.

Parameters:

- Motor idle current $I_{\text{motor_idle}} = 0.06 \text{ mA}$

- Motor active current $I_{\text{motor_active}} = 2440 \text{ mA}$ (average during sweep)
- Motor duration $T_{\text{motor}} \approx 5 \text{ s}$ per sweep
- Update interval $T_{\text{interval}} = 1 \text{ h} = 3600 \text{ s}$

$$\begin{aligned}
 I_{\text{avg_servo}} &= I_{\text{motor_idle}} + \frac{I_{\text{motor_active}} T_{\text{motor}}}{T_{\text{interval}}} \\
 &= 0.06 \text{ mA} + \frac{(2440 \text{ mA})(5 \text{ s})}{3600 \text{ s}} \\
 &= 0.06 \text{ mA} + 3.39 \text{ mA} \approx 3.45 \text{ mA}.
 \end{aligned}$$

8.5 Battery Lifetime Estimation

Each battery-powered domain uses a standard 2000 mA h cell. A safety/efficiency derating factor of 0.85 is applied to account for non-ideal discharge, self-discharge, and conversion losses, giving a usable capacity of

$$C_{\text{usable}} = 2000 \text{ mA h} \times 0.85 = 1700 \text{ mA h}.$$

Transmitter Node Lifetime.

$$\begin{aligned}
 t_{\text{tx}} &= \frac{C_{\text{usable}}}{I_{\text{avg_tx}}} = \frac{1700 \text{ mA h}}{1.356 \text{ mA}} \approx 1253 \text{ h} \\
 &\approx 52 \text{ d} \quad (\text{about 1.7 months}).
 \end{aligned}$$

Servo Battery Lifetime.

$$\begin{aligned}
 t_{\text{servo}} &= \frac{C_{\text{usable}}}{I_{\text{avg_servo}}} = \frac{1700 \text{ mA h}}{3.45 \text{ mA}} \approx 492.7 \text{ h} \\
 &\approx 20.5 \text{ d}.
 \end{aligned}$$

Receiver Node. At $I_{\text{base}} = 470 \text{ mA}$ continuous, the same derated 1700 mA h cell lasts only $1700/470 \approx 3.6 \text{ h}$, so the receiver remains wall powered.

8.6 Summary

Table 1 consolidates the measured currents, modeled averages, and estimated lifetimes for the three power domains.

Table 1. Power and lifetime summary by subsystem.

Subsystem	Avg. current	Battery	Lifetime
Transmitter node (LoRa)	1.356 mA	2000 mA h LiPo	~52 d
Servo motor (isolated)	3.45 mA	5 V to 7 V pack, 2000 mA h	~20.5 d
SmartHub receiver	470 mA	Wall power	continuous

The analysis validates the decision to separate the system into three isolated power domains. Keeping the SmartHub on wall power avoids the 3.6 h collapse that a battery-only design would suffer, the isolated servo supply shields the ESP32 from the motor’s 8.2 A startup spikes, and deep-sleep operation on the remote transmitter allows it to run for roughly 52 d on a single 2000 mA h LiPo cell.

9 Conclusions

The IoT Connected SmartHub Dashboard successfully demonstrates the integration of multiple IoT technologies into a single user-friendly smart desk system. The project combines LoRa, Bluetooth Low Energy (BLE), and Wi-Fi communication with environmental sensing, web server functionality, graphical dashboard rendering, and camera-based gesture recognition using ESP32 microcontrollers.

Testing results demonstrated reliable LoRa communication, successful BLE device scanning, stable Wi-Fi interaction, and accurate hand gesture recognition performance. Power analysis and battery lifetime measurements also validated the use of isolated power domains to improve system stability, communication reliability, and power efficiency.

Overall, the SmartHub Dashboard provides students and everyday users with convenient access to useful real-time information through a centralized and interactive platform while demonstrating the practical implementation of modern IoT communication, embedded systems, and edge machine learning technologies.

10 Recommendations

- Improve final subsystem integration and reduce communication delays between ESP32 devices.
- Continue optimizing the hand gesture recognition model to reduce false detections and improve overall accuracy.
- Improve power management and battery efficiency for more portable system operation.
- Add additional environmental sensors to expand monitoring capabilities.

- Improve the SmartHub user interface with smoother animations and improved screen organization.
- Expand LoRa communication testing in more realistic indoor and outdoor environments.
- Implement more secure communication methods for Wi-Fi and BLE connections.
- Improve hardware organization using a custom 3D-printed enclosure and mounting system.
- Add cloud connectivity for remote monitoring, synchronization, and data storage.
- Continue long-term reliability and stress testing under continuous system operation.

Appendix A System Architecture

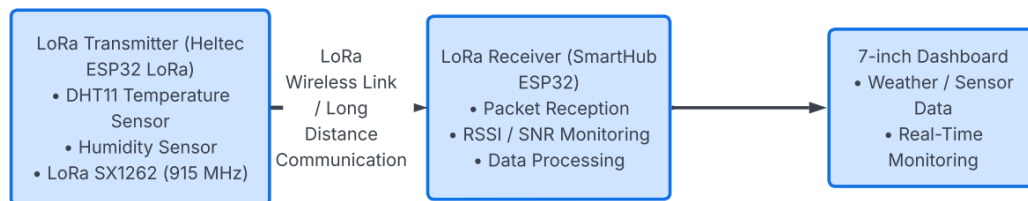


Figure 5. LoRa communication block diagram.

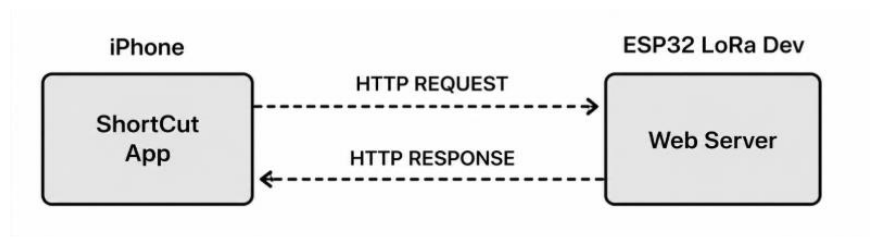


Figure 6. Wi-Fi and BLE architecture diagram.

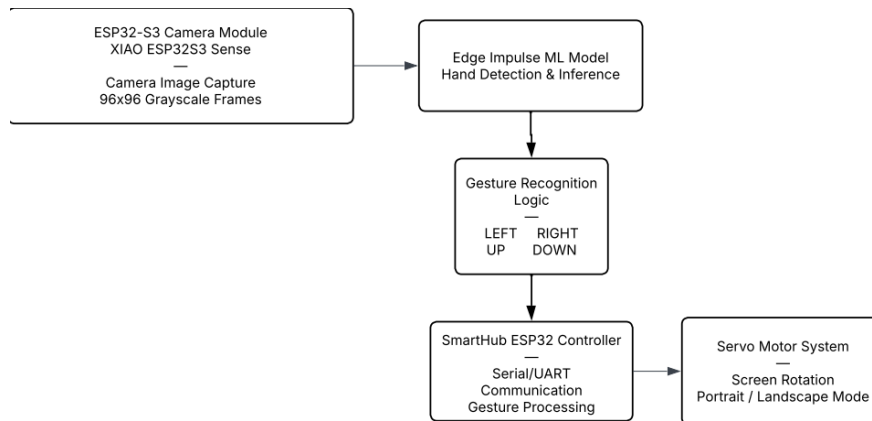


Figure 7. Camera gesture system architecture.

Appendix B Circuit Diagrams

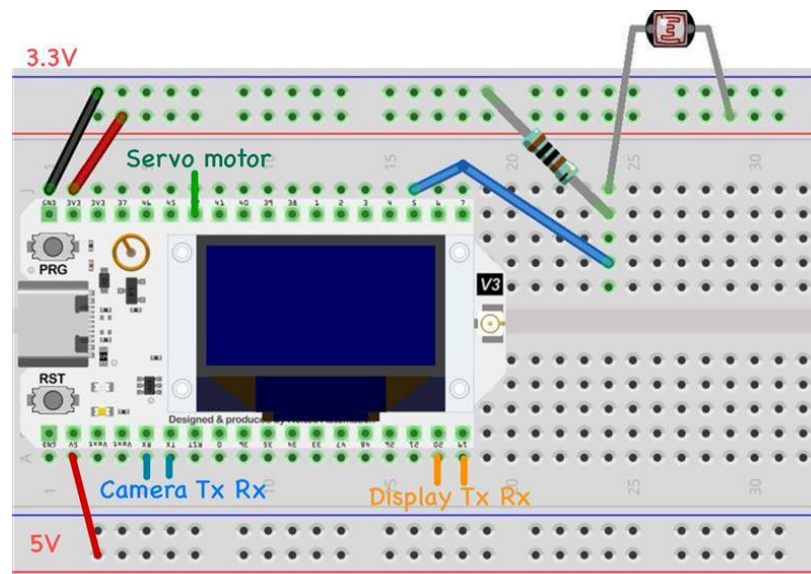


Figure 8. Main Hub ESP32 pin connections. Camera and display relate to the UART pins.

Appendix C UI Screenshots

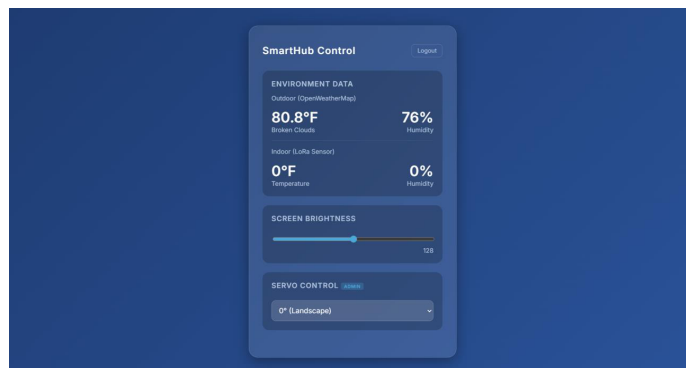


Figure 9. Web server interface.



Figure 10. Gesture recognition demo capturing the four directional classes (RIGHT, LEFT, UP, DOWN).

Appendix D Bill of Materials

Table 2. Bill of Materials

Item	Details	Cost (USD)
Screen	7-inch 800 × 480 ESP32 Screen	\$53.49
Camera	Xiao ESP32-S3 Camera	\$21.83
Motor	20 kg RC Servo (High Torque)	\$14.00
Photoresistor	In-house (parts room)	–
Humidity sensor	In-house (parts room)	–
Temperature sensor	In-house (parts room)	–
Total		\$89.32
Cost per person		\$29.73